Shuaicheng Zhang

PHD CANDIDATE · COMPUTER SCIENCE DEPARTMENT OF VIRGINIA TECH

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Research Interest

My research develops **efficient inference methods** and **data strategies** for foundation models reasoning over structured knowledge. I design **scalable architectures** that integrate graph-structured context with language models (handling billions of edges), build **training data pipelines** from structured sources (knowledge graphs, temporal networks), and develop systems that adapt to evolving data distributions. My work emphasizes **efficient multi-modal reasoning**, **retrieval-augmented generation** for structured domains, and **training recipes** for graph-grounded tasks, with applications in predictive modeling, financial intelligence, database query understanding, and scientific discovery.

Education

Virginia Tech — *Blacksburg, VA*
PHD CANDIDATE COMPUTER SCIENCE — *August 2020 - present*
- Advisor: Dr. Dawei Zhou

MIT CSAIL — *Cambridge, MA*
VISITING PHD STUDENT — *May 2023 - Aug 2023*
- Advisor: Dr. Julian Shun

University of Maryland, College Park — *College Park, MD*
MS TELECOMMUNICATIONS — *August 2018 - May 2020*
- Advisor: Dr. Michael Dellomo

Virginia Tech — *Blacksburg, VA*
BS COMPUTER SCIENCE — *August 2013 - Dec 2017*
- Minors in Mathematics
- Advisor: Dr. T.M. Murali

Industrial Experience

Oak Ridge National Lab — *Oak Ridge, TN, USA*
RESEARCH INTERN — *Spring 2026*
- Developing **Graph RAG systems** for scientific discovery: building NLP pipelines to extract entities, relations, and claims from neuroscience literature, creating instruction-tuning datasets for graph-grounded reasoning and hypothesis generation
- Designing **LLM training data workflows** at scale: deploying automated knowledge extraction pipelines on HPC systems, developing data recipes combining graph context with retrieved literature, and establishing evaluation protocols for scientific LLM outputs

Gray Systems Lab, Microsoft Research — *Redmond, WA, USA*
RESEARCH INTERN — *Summer 2025*
- Developed **WorkloadGPT**, a multimodal foundation model integrating GNNs and LLMs to jointly reason over structured knowledge (SQL queries and execution plans) and temporal telemetry data for intelligent database workload identification and optimization across Azure services
- Built **data pipelines** processing millions of SQL workloads, achieving SOTA performance on workload identification task

CSAIL, MIT-IBM Watson AI Lab — *Cambridge, MA, USA*
RESEARCH INTERN — *Summer 2023*
- Developed **H^{2}GB** benchmark for evaluating GNNs on heterogeneous and **H^{2}G-former** for a modular graph transformer with capability of large scale graph learning, winning **KDD 2025 Best Paper Award**
- Designed **UnifiedGT**, a unified graph transformer framework, bridging effective and efficient approaches for scalable systems handling billions of edges, published in IEEE BigData 2024

Natural Language Processing Team, Deloitte AI Center of Excellence *New York, NY, USA*

NLP Research Intern *Summer 2021*

- Designed **open relation extraction system** based on GPT-3 for analyzing important events and relationships from audit files, helping auditors better identify critical patterns and connections in financial documents

Trust Team, Hundsun Technologies Inc. *Hangzhou, China*

Software Development Engineer Intern *Summer 2017*

- Designed and implemented high-performance multithreaded PDF converter with concurrent processing workflows for financial document processing at scale

Preprints & Publications

Scientific Hypothesis Generation and Validation: Methods, Datasets, and Future Directions ——Adithya Kulkarni, Fatimah Alotaibi, Xinyue Zeng, Longfeng Wu, Tong Zeng, Barry Menglong Yao, Minqian Liu, **Shuaicheng Zhang**, Lifu Huang, Dawei Zhou. In Arxiv, Under review

CAPTAIN: Conformal-Prediction-Based Multi-Source Time-Series Forecasting ——**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. Under review of **TMLR**'2025.

PDMBench: A Standardized and Multi-Perspective Benchmark for Predictive Maintenance on Multi-Sensor Industrial Time-Series Data ——**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. Under review of **ICLR**'2025.

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HeroFilter: Adaptive Spectral Graph Filter for Varying Heterophilic Relations ——**Shuaicheng Zhang**, Haohui Wang, Junhong Ling, Xiaojie Guo, Yada Zhu, Si Zhang, Dongqi Fu, Dawei Zhou.(*Equal Contribution) In preceedings of the (**NeurIPs**)'2025.

When Heterophily Meets Heterogeneity: New Graph Benchmarks and Effective Methods ——Junhong Lin, Xiaojie Guo , **Shuaicheng Zhang**, Yada Zhu, Julian Shun. In Preceedings of the ACM **KDD**'2025. **Best Paper Award**

MentorPDM: Learning Data-Driven Curriculum for Multi-Modal Predictive Maintenance ——**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. In Preceedings of the ACM **KDD**'2025.

UnifiedGT: Towards a Universal Framework of Transformers in Large-Scale Graph Learning ——Junhong Lin, Xiaojie Guo , **Shuaicheng Zhang**, Yada Zhu, Dawei Zhou, Julian Shun. In Preceedings of the **IEEE BigData** 2025. **Patent Pending**.

TGEditor: Task-Guided Graph Editing for Augmenting Temporal Financial Transaction Networks ——**Shuaicheng Zhang**, Yada Zhu, Dawei Zhou. In Proceedings of the ACM International Conference on AI in Finance(**ICAIF**)'2023(**oral**). **Patent Pending, Deployed at Wellsfargo**.

Personalized Federated Learning under Mixture of Distributions ——**Shuaicheng Zhang***, Yue Wu*, Yanchi Liu, Quanquan Gu, Dawei Zhou, Haifeng Chen, Wei Cheng. (*Equal Contribution) In Proceedings of International Conference on Machine Learning(**ICML**)'2023.

Extracting Temporal Event Relation with Syntax-guided Graph Transformer ——**Shuaicheng Zhang**, Ning Qiang, Lifu Huang. In Findings of North American Chapter of the Association for Computational Linguistics(**NAACL**)'2022.

Awards & Honors

2025	**KDD Best Paper Award in Benchmark and Dataset Track**, KDD 2025
2025	**KDD Outstanding Reviewer (Top 10%)**, KDD 2025
2024	**CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
2023	**CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
2022	**CIKM Travel Award**, CIKM 2022
2021	**Virginia Tech Pratt Fellowship**, Virginia Tech Department of Computer Science

Academic Service

Conference Reviewer

2026 **AAAi 2026, ICLR2026**, Reviewer, Program Committee
2025 **KDD 2025, ICML 2025, ICLR 2025, Neurips 2025**, Reviewer, Program Committee
2024 **KDD 2024, ICML 2024**, Reviewer, Program Committee
2023 **KDD 2023, ICDM 2023, WWW 2023**, Reviewer
2022 **SIGIR 2022, KDD 2022, NLPCC 2022, WWW 2023**, Reviewer
2021 **SDM 2022, WWW 2022, CIKM 2022**, Subreviewer
2021 **NLPCC 2021**, Reviewer

Organizing committee

2022 **The 1st Workshop on Trustworthy Learning on Graphs (TrustLOG) at The Conference on Information and Knowledge Management (CIKM)'2022**, Publicity Chair *Atlanta*

Skills

Statistics Advanced statistical analysis, inference, factor, regression analyses, hypothesis testing, hierarchical modeling

Machine Learning and Deep Learning TensorFlow, Keras, Pytorch, Scikit-Learn, XGBoost, Pandas, Numpy, Scipy, NLTK, NetworkX

Programming Python, Java, C, Matlab, Shell, MySQL, HTML

Basics Linux, Git, LaTex, Docker

Teaching Experience

Spring 22 **CS 5984 Computational Social Science**, Teaching Assistant
Fall 21 **CS 5984 Natural Language Processing**, Teaching Assistant
Spring 21 **CS 2505 Computer Architecture I**, Teaching Assistant
Fall 20 **CS 3214 Operating System**, Teaching Assistant